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Submission: Phase 2 Report

1. Introduction

For this 2nd homework I made use of the (region_2.csv) dataset provided to me by Professor Rozsa Zaruba. This dataset she provided to me is a subset of the Global Terrorism Database and it contains over 10,000 incidents of terrorist attacks from 1970 to 2020 (inclusive of both years) in the Central America and the Caribbean regions. I analyzed the dataset using Python, GeoPandas, Folium along with various other time series tools.

The main objectives were to clean the data while building the relevant geospatial visualizations and then study the distribution of attack types while exploring temporal trends in the count of terrorist attack incidents over the years (5 decades from 1970 to 2020).

2. Data preparation

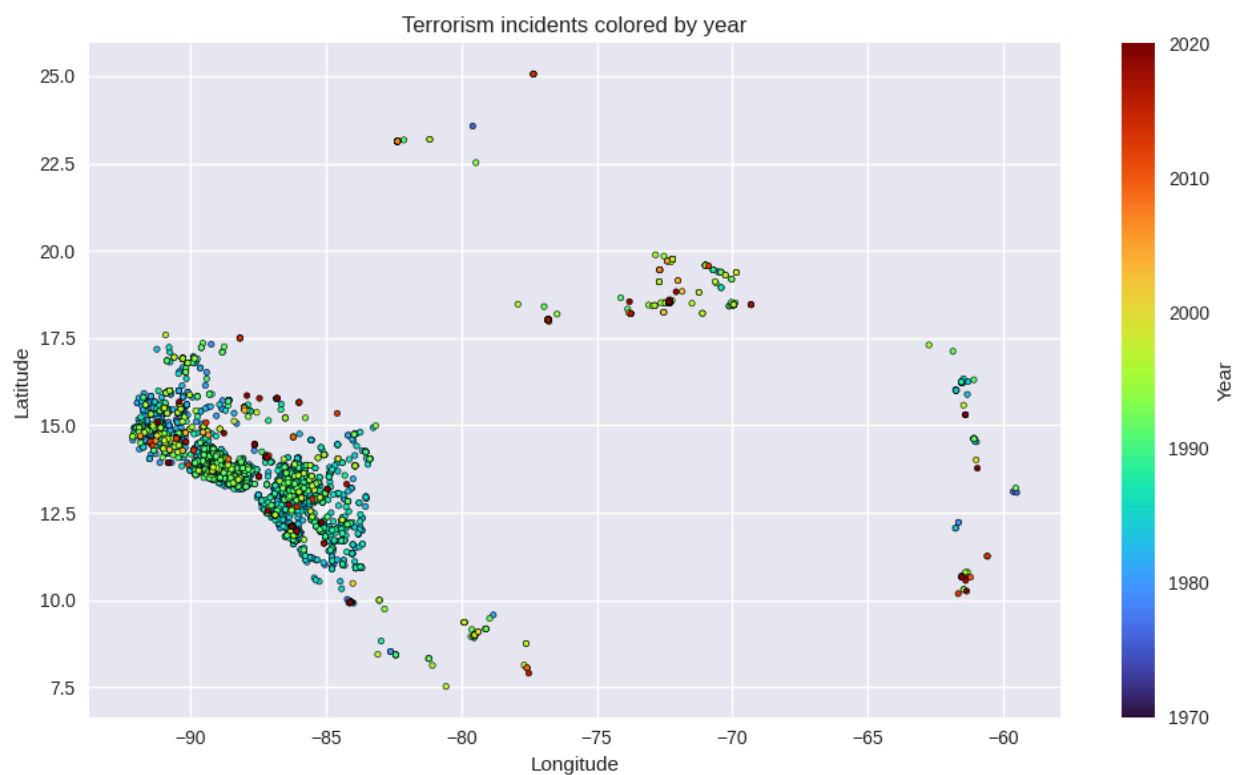
I first loaded the provided (region_2.csv) file into a pandas DataFrame and inspected its shape from where I found that it had 10,386 rows, as well as 135 columns. I selected only the variables required by the assignment: (eventid, iyear, imonth, iday, country_txt, region_txt, city, latitude, longitude, attacktype1_txt, nkill and nwound). I had to make sure that the coordinates were correct for mapping which was why I decided to remove all 1,000 rows with missing values for longitude and latitude. In total I kept 9,386 incidents (rows) for the analysis.

Using the cleaned table, I created a GeoDataFrame by converting each terror incident's longitude and latitude into a Point geometry in WGS84 (EPSG:4326) format. I displayed the first five rows of this GeoDataFrame to confirm that the geographic information and selected attributes looked correct. All incidents were in the Central America and

Caribbean region. Especially in cities such as Santo Domingo (Dominican Republic) and Guatemala City (Guatemala) along with the attack types such as Assassination and Hostage Taking.

I then showed summary statistics for the key variables (iyear, imonth, iday, nkill and nwound) which showed that the data span from 1970 to 2020 (including both years) with a median year of 1984. Most of the incidents took place in mid year months and mid month days. Long tailed fatality distributions had up to 300 people killed along with 130 people wounded (in the most extreme cases).

After that I made use of the inbuilt plotting of GeoPandas to create a simple scatter plot of all incidents colored by year. This plot already revealed that nearly all points lie within a narrow latitude and longitude band covering Central America. The density of incidents is highest in a corridor from southern Mexico through Guatemala, El Salvador, Honduras and Nicaragua and also in smaller clusters in the Caribbean.



As we can see from this plot that incidents are heavily clustered along the Central American and the Caribbean regions. The color scale shows that terrorist incidents in the 1970s are not very frequent but later years fill in the same regions with many more points. The plot conveyed to me how the conflicts intensified and became worse over time while being focused in mostly the same set of locations rather than spreading evenly across the broader region.

3. Geospatial visualization

3.1 Base map with incident markers

To produce an interactive base map I used Folium and centered the map at the mean latitude and longitude of all incidents. I then iterated over the GeoDataFrame and added a small (CircleMarker) for each event with a red marker and a popup showing the city, country and year. The HTML map that I got as an output (terrorism_base_map.html) clearly shows dense clusters of incidents around San Salvador, Guatemala City and Managua. Additional concentrations are present in parts of Honduras as well as Nicaragua. This confirms the fact that terrorist activities in this dataset were highly concentrated in a few major areas rather than being evenly spread out.

3.2 Choropleth map by country

For the choropleth map I first arranged the data in various groups by (country_txt) to compute the number of incidents that took place per country. The top 10 countries are El Salvador (4,846 incidents), Guatemala (1,962), Nicaragua (1,579), Honduras (326), Haiti (223), Panama (127), Dominican Republic (89), Costa Rica (69), Guadeloupe (39) and lastly Jamaica (34 incidents).

```

...
Top 10 countries by incident count (for sanity check):
rank  country_txt  incident_count
0     1      El Salvador      4846
1     2      Guatemala      1962
2     3      Nicaragua      1579
3     4      Honduras       326
4     5       Haiti        223
5     6       Panama       127
6     7  Dominican Republic    89
7     8      Costa Rica       69
8     9      Guadeloupe       39
9    10       Jamaica       34
Using cached world_countries.json
Saved: terrorism_choropleth_map.html

```

After that I joined these counts to an external GeoJSON of world country boundaries (world_countries.json) and used Folium's Choropleth layer to make the visualizations of the counts of terrorist attack incidents per country. On the resulting map El Salvador appears in the darkest color (dark red), followed by Guatemala and Nicaragua (light orange). The remaining countries (Cuba, Dominican Republic, Belize, Jamaica and Costa Rica among others) are shaded more lightly (shades of yellow) which shows how strongly the regional share of the incidents is present largely in just 3 nations.

3.3 Heatmap of incidents

To emphasize spatial density rather than country totals I made a Folium heatmap using all incident coordinates. Each incident contributed a point to the HeatMap layer. This was provided on a dark colored basemap with moderate radius and blurred settings. The heatmap supports the earlier conclusions that the hottest areas link to El Salvador, western and central Guatemala and the Pacific side of Nicaragua with much of the broader region having very little terrorist activity.

4. Attack type analysis

4.1 Frequency counts

For attack type analysis I created a frequency table for the (attack-type1_txt) column using (value_counts). The results are:

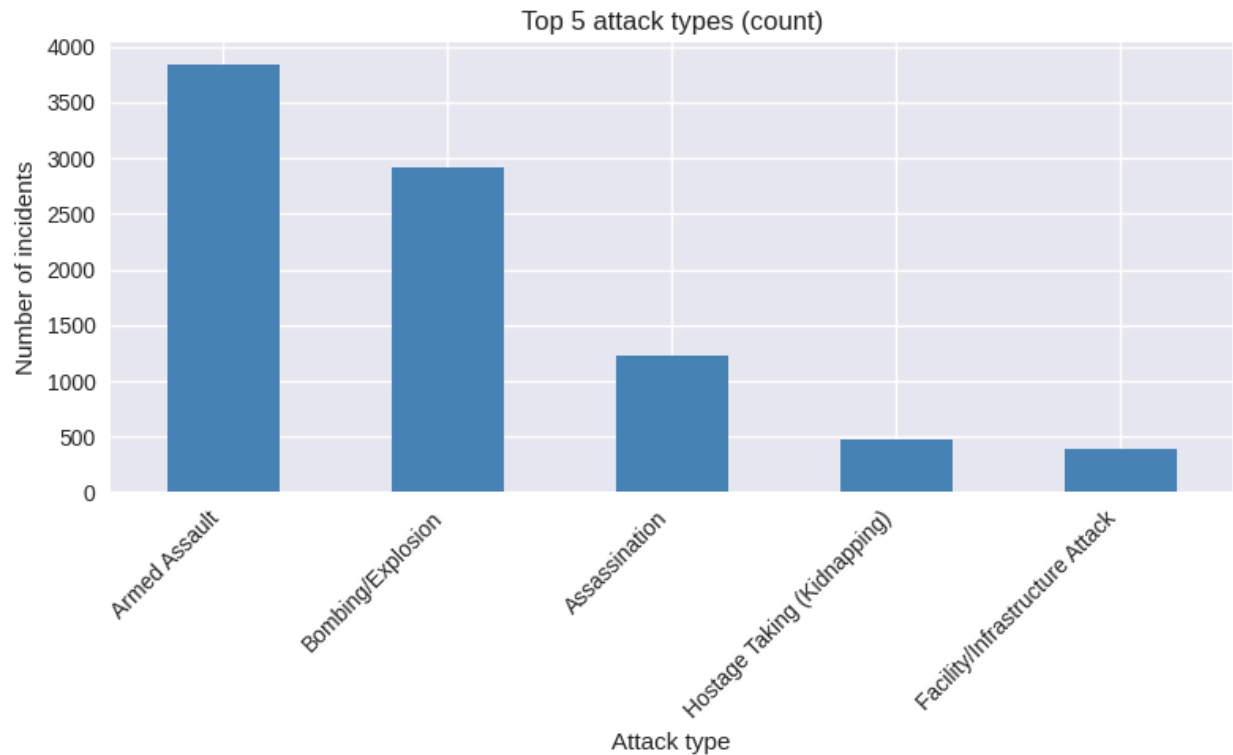
- Armed Assault: 3,842 incidents
- Bombing/Explosion: 2,912 incidents
- Assassination: 1,232 incidents

- Hostage Taking (Kidnapping): 475 incidents
- Facility/Infrastructure Attack: 390 incidents
- Unknown: 310 incidents
- Hostage Taking (Barricade Incident): 181 incidents
- Hijacking: 25 incidents
- Unarmed Assault: 19 incidents

```
... Attack type frequencies:
attacktype1_txt
Armed Assault          3842
Bombing/Explosion      2912
Assassination          1232
Hostage Taking (Kidnapping)  475
Facility/Infrastructure Attack  390
Unknown                310
Hostage Taking (Barricade Incident)  181
Hijacking               25
Unarmed Assault         19
Name: count, dtype: int64
```

These counts show that armed assaults and Bombings are by far the two major methods of terror attacks. They are together responsible for the majority of recorded incidents.

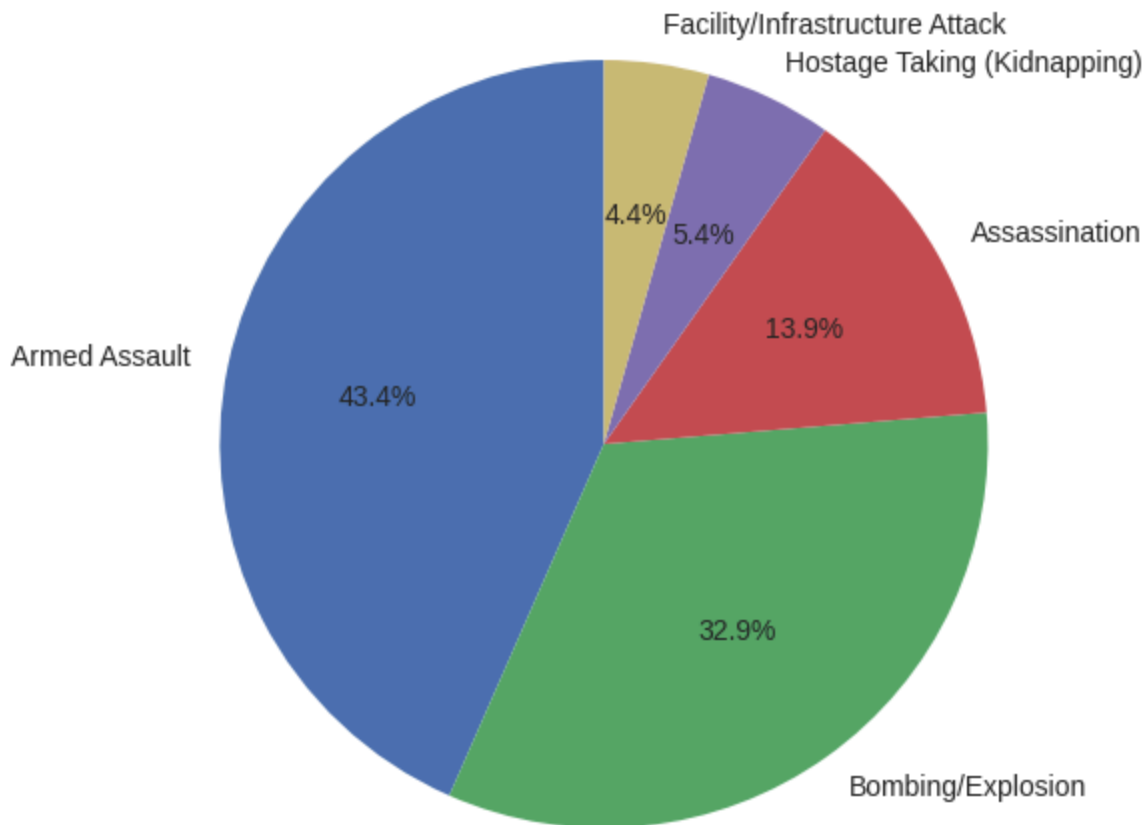
4.2 Bar Chart and Pie Chart



Bar Chart

As we can see clearly from the bar chart below armed assaults are the most common type of terrorist attack in this region close to 4,000 recorded incidents which was followed by Bombings (Explosions) around 3,000 terrorist events. Assassination appears as a smaller third category while Hostage Taking (Kidnapping) and Facility/Infrastructure Attacks have much lower counts and make up less frequent tactics. This suggests that most of the incidents in Central America and the Caribbean involved either armed attacks or explosions with attacks on facilities having a very small role.

Top 5 attack types (share of incidents)



Pie Charts (share of incidents)

The pie chart represents basically the same information in percentage terms and shows how dominant the first two categories really are. Armed Assault on its own makes up for about 43.4% of all incidents. Bombings (Explosions) contributes 32.9% of all such terrorist activities. Assassination adds up to 13.9% of incidents. Each of the remaining categories such as Hostage Taking (Kidnapping) and Facility/Infrastructure Attack, contributes only around 4 to 5% of the total share of terror events.

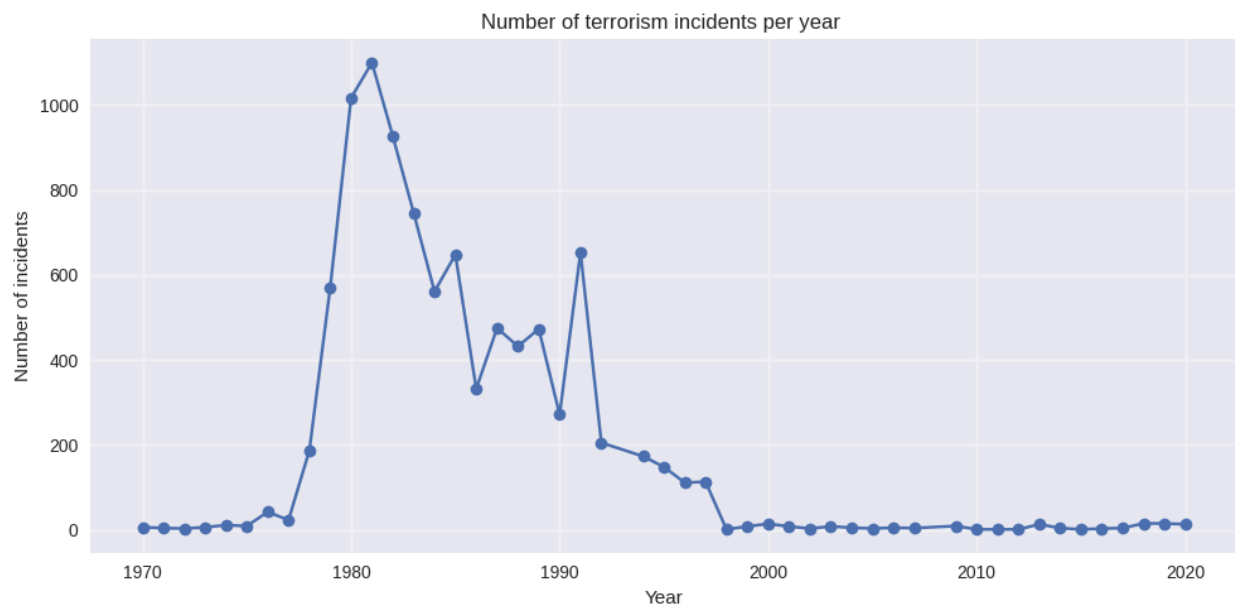
This visualization implies that armed groups in this region primarily relied on gunfire and explosives while using assassinations and infrastructure related attack operations as secondary tools instead of using such tactics as their primary strategy.

5. Time series analysis

5.1 Yearly incident counts

For the time series component I grouped incidents by (iyear) and counted how many events occurred in each year. The first 10 years show very low terrorist activity starting at 6 incidents in 1970, 4 in 1971, 3 in 1972 and then eventually increasing to 11 incidents in 1974, 9 in 1975, 42 in 1976, 23 in 1977, 187 in 1978 and 571 terror events in 1979.

```
Yearly incidents (first 10 rows):  
iyear  
1970      6  
1971      4  
1972      3  
1973      6  
1974     11  
1975      9  
1976     42  
1977     23  
1978    187  
1979    571  
dtype: int64
```



I plotted these yearly counts as a line chart titled “Number of terrorism incidents per year”. The plot shows a huge rise in incidents from the late 1970s into the early 1980s with a peak of around 1,100 incidents around 1981. It is followed by a gradual decline in the mid 1980s and such terrorist incidents again drop further in the subsequent decades of the 1990s and the 2000s.

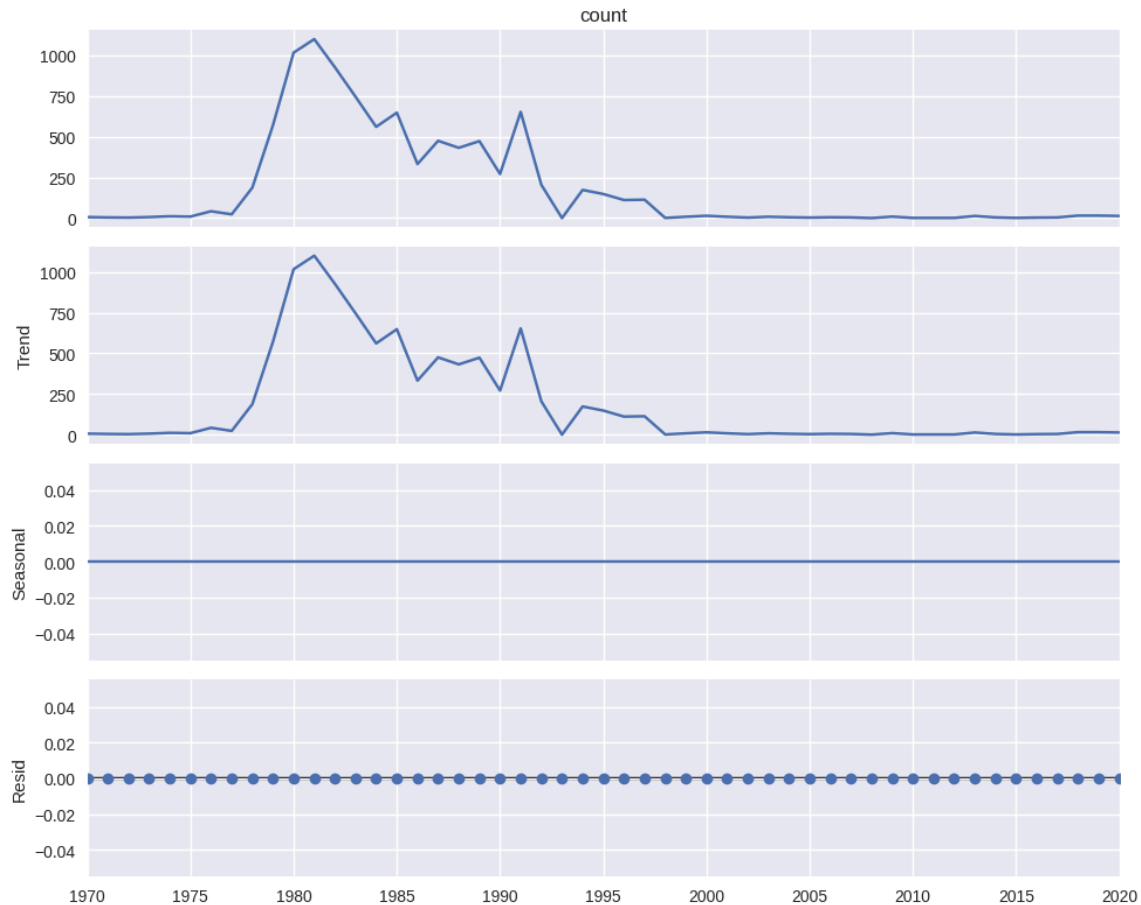
5.2 Decomposition and interpretation

In order to examine the underlying patterns and trends (structure) I converted the yearly counts into an annual time series with a `DateTimeIndex` (one point per January 1) and applied an additive seasonal decomposition with a period of 1.

The decomposition produced four panels: the observed series (count), a smooth trend component, a seasonal component, and residuals (Resid). The basic statistics of yearly counts show an average (mean) of approximately 184 incidents per year, a very large standard deviation (std: around 296), a minimum (min) of 0, a median (50th percentile) of 13 alongside a maximum of 1,100 incidents.

```
...
=== Basic stats of yearly incidents ===
count      51.000000
mean       184.039216
std        296.558794
min         0.000000
25%         4.000000
50%        13.000000
75%       238.000000
max       1100.000000
Name: count, dtype: float64
```

The trend part closely follows the overall shape of the insights observed initially as there is an increase from near zero in the early 1970s to a peak around the early 1980s after which there is a decline toward low levels after the mid 1990s. The seasonal component is basically flat at zero because annual data at this resolution do not exhibit any major changes within a single year.



The residual (Resid) series is very small and close to zero which implies that there is little remaining unexplained structure (patterns or trends) once the big picture trends are taken into account. Most of the variability is primarily because of the long term rise and fall in the frequency (intensity or number of terror attacks over a time period) of terrorism rather than regular cyclical effects.

These results suggest that terrorism in this region shifted from irregular incidents in the early 1970s to extremely tense conflicts in the late 1970s and early 1980s before declining in a rapid manner and eventually stabilizing at lower levels towards the end of the period when viewed from an overall perspective.

6. Conclusion

What I concluded from this analysis is that terrorist incidents in the Central America and Caribbean region are geographically centered in a few countries and urban centers which are largely dominated by El Salvador, Guatemala and Nicaragua while including cities such as San Salvador, Guatemala City and Managua. The most common attack types are Armed Assault and Bombing/Explosion all of which together account for more than 75% of the incidents. Assassination and kidnapping (hostage taking) events form smaller but not insignificant shares.

This time series analysis ultimately tells us that there was a serious escalation of violence in the late 1970s and early 1980s after which it was followed by a long decline to low levels by the 2000s as there were no strong seasonal patterns. Basically we infer from all of this that the main driver of change was the rise and eventual resolution of conflicts in this region rather than regular yearly cycles.